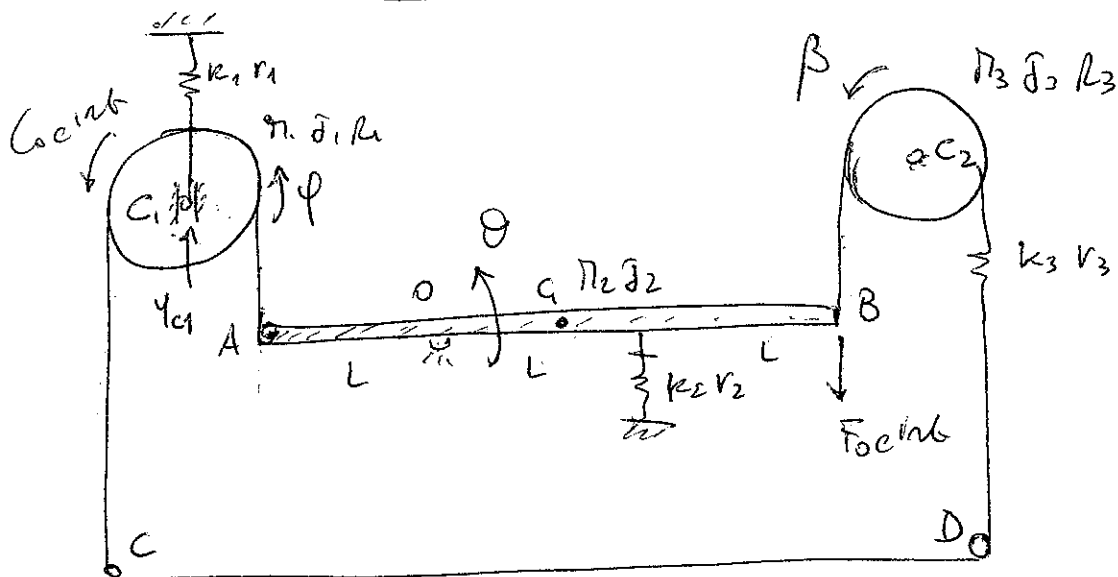


Solutione 25 febbraio 2008



$$x = \begin{Bmatrix} \phi \\ \theta \end{Bmatrix}$$

$$E_c = \frac{1}{2} m_1 \dot{y}_1^2 + \frac{1}{2} J_1 \dot{\phi}^2 + \frac{1}{2} m_2 \dot{y}_2^2 + \frac{1}{2} J_2 \dot{\theta}^2 + \frac{1}{2} J_3 \dot{\beta}^2$$

$$(m_F) = \text{diag} (m_1, J_1, m_2, J_2, J_3)$$

$$\begin{cases} y_1 = -L\theta - R_1\phi \\ y_2 = \frac{L}{2}\theta \\ \beta = -\frac{2L}{R_3}\theta \end{cases} \quad (J_m) = \begin{bmatrix} -R_1 & -L \\ 1 & 0 \\ 0 & L/2 \\ 0 & 1 \\ 0 & -\frac{2L}{R_3} \end{bmatrix}$$

$$U_{El} = \frac{1}{2} k_1 \Delta l_1^2 + \frac{1}{2} k_2 \Delta l_2^2 + \frac{1}{2} k_3 \Delta l_3^2 \quad (k_F) = \text{diag} (k_1, k_2, k_3)$$

$$\begin{cases} \Delta l_1 = -y_1 = L\theta + R_1\phi \\ \Delta l_2 = L\theta \\ \Delta l_3 = -2L\theta - L\theta - 2R_1\phi \end{cases} \quad (J_{El}) = \begin{bmatrix} +R_1 & L \\ 0 & L \\ -2R_1 & -3L \end{bmatrix}$$

$$V_G = m_1 g h_{G1} + m_2 g h_{G2} = m_1 g (-L \sin \theta - R_1 \phi) + m_2 g \frac{L}{2} \sin \theta$$

$$\frac{\partial^2 V_G}{\partial \theta^2} = (m_1 g L - m_2 g \frac{L}{2}) \sin \theta \rightarrow \frac{\partial^2 V_G}{\partial \theta^2} \Big|_0 = 0$$

$$\frac{\partial^2 V_G}{\partial \phi^2} = 0 \quad \frac{\partial^2 V_G}{\partial \phi \partial \theta} = 0 \rightarrow [k_G] = \emptyset$$

$$\delta W = C_{oeint} \delta \phi + \vec{F}_{oeint} \times \delta y_B = C_{oeint} \delta \phi + \vec{F}_{oeint} \cdot 2L \delta \theta$$

$$Q_c = \begin{Bmatrix} 1 & 0 \end{Bmatrix} C_{oeint} \quad ; \quad Q_F = \begin{Bmatrix} 0 & -2L \end{Bmatrix} F_{oeint}$$